

Venn diagrams

Conditional probability

Independent vs. dependent
events

Multiplication rules

Conditional probability and
diagnostic testing

L12: More Probability

Learning Objectives

Venn diagrams

Conditional probability

Independent vs. dependent
events

Multiplication rules

Conditional probability and
diagnostic testing

- ▶ More on Venn diagrams
- ▶ General addition rule for probability
- ▶ Conditional probability
- ▶ Determine whether two events are independent or dependent
- ▶ General multiplication rule for probability

Venn diagrams

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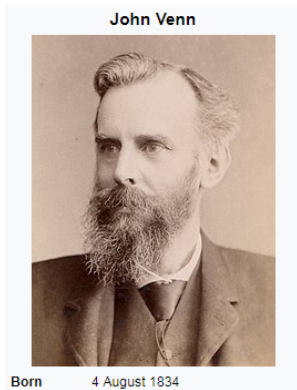
Multiplication rules

Conditional probability and
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Venn diagrams

Venn diagrams

Popularized in a paper in 1880



Venn diagrams

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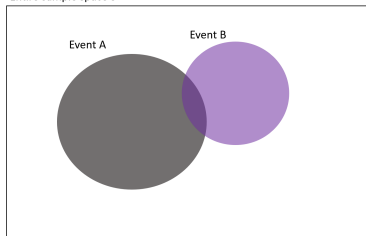
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Venn diagrams

A basic Venn diagram

Entire sample space S



- ▶ Venn diagrams generally consist of at least two events
- ▶ When we draw a rectangle around the diagram this is the sample space. All the probabilities inside must sum to 1.
- ▶ When you make a Venn diagram label the events using capital letters to represent the random variables

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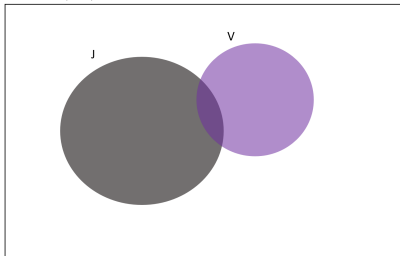
Venn diagrams

Suppose you had access to survey data about vaping and JUUL-related advertisements.

You could define the following random variables: J is the event “seen ad for JUUL” V is the event “vaped in the last 30 days”

A basic Venn diagram

Entire sample space S



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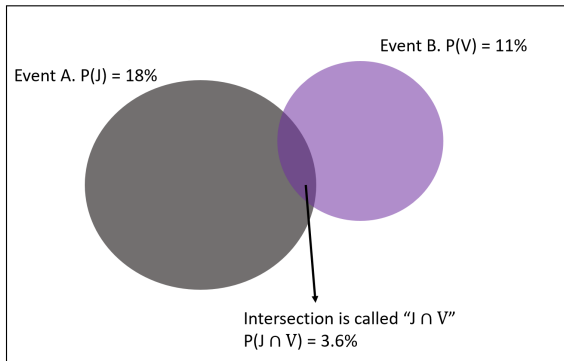
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Now let's say we do a survey - we can add the percents to the diagram:

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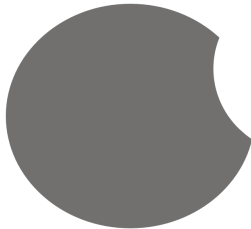
Conditional probability and
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what percent of the individuals saw an ad for JUUL and do not vape?

Venn diagram

What percent of individuals saw an ad for JUUL and do not vape?

This percent is represented by this area:



How would we write this as a probability statement? What percentage is this?

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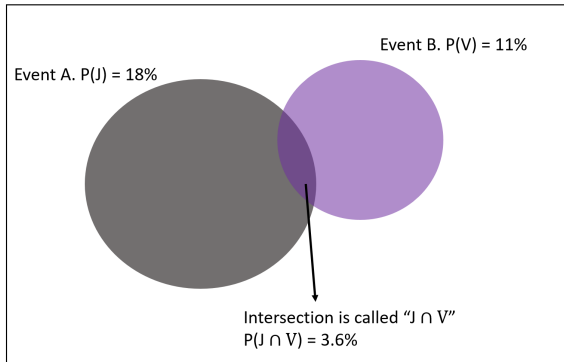
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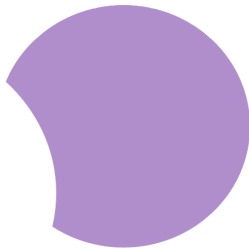
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What percent of individuals vape but have not seen an ad for JUUL?

Venn diagram

What percent of individuals vape but have not seen an ad for JUUL? This percent is represented by this area:



How would we write this as a probability statement? What percentage is this?

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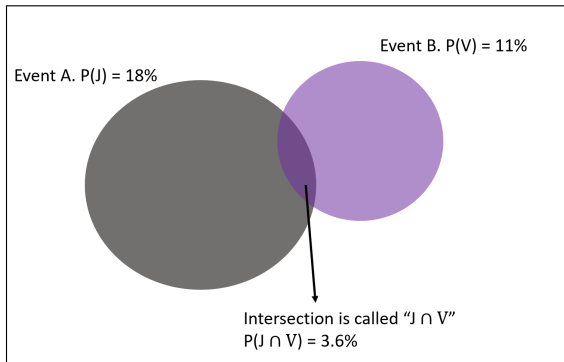
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What percent of individuals vape AND have seen an ad for JUUL? What is this called?

Venn diagram

Venn diagrams

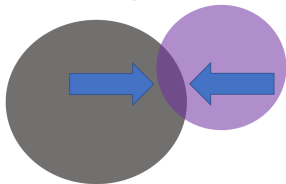
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**What percent of individuals have seen an
ad for JUUL and vape?**



The “and” represents the intersection, the
area where both events have occurred:

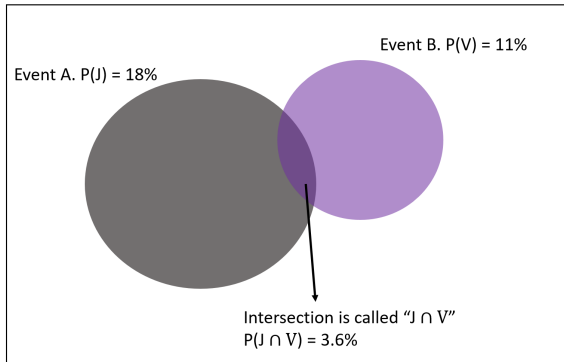
$$P(J \cap V) = 3.6\%$$

3.6% of individuals have seen an ad for
JUUL and have vaped in past 30 days

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What percent of individuals vape OR seen an ad for JUUL? What is this called?

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The **union** is all events that have either of the outcomes - all the shaded parts of this diagram are part of the union.

$$P(J \text{ or } V)$$

$$P(J \cup V)$$

$$= P(J) + P(V) - P(J \cap V) \text{ this is the general rule for addition}$$

$$= 0.18 + 0.11 - 0.036$$

$$= 0.254 = 25.4\%$$

For any two events A and B , $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

- ▶ Why do we subtract off $P(A \cap B)$?
- ▶ This formula simplifies to $P(A \cup B) = P(A) + P(B)$ when A and B are **disjoint**.
- ▶ What does the Venn diagram look like for disjoint events?

Decomposition of a probability

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For any two events A and B , $P(A) = P(A \cap B) + P(A \cap \bar{B})$

Which pieces of our diagram does this indicate?

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Statistics is everywhere

You may have heard people talk about the probability of dying from a bee sting being greater than the probability of being killed by a shark. As in this article, published during “shark week”



ENTERTAINMENT

Shark week: You're way more likely to die from these than a shark attack

Here are seven things more likely to kill you than sharks.

From the article:

- ▶ Your odds of a dog killing you are 1 in 112,400
- ▶ Your odds of dying from one of these insects touching you is 1 in 63,225
- ▶ Your odds of dying from a fired gun are 1 in 6,905
- ▶ Your odds of dying from lightning are 1 in 161,856
- ▶ Your odds of dying while in a car are 1 in 114

compared to about 1 in 3,748,067 for a shark attack.

It's worth noting that these are probabilities (risks) **NOT** odds in the statistical sense

People use these probabilities to argue that swimming in the ocean is safe. Why do conditional probabilities make this argument questionable?

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Conditional probability



RISK OF PREDIABETES:
1 IN 3 ADULTS

RISK OF SHARK ATTACK:
1 IN 11.5 MILLION

**LIFE DOESN'T ALWAYS
GIVE YOU TIME TO
CHANGE THE OUTCOME.**

PREDIABETES DOES.

TAKE THE RISK TEST TODAY AT
DoIHavePrediabetes.org

ad COUNCIL
AMA AMERICAN MEDICAL ASSOCIATION
CDC CENTERS FOR DISEASE CONTROL AND PREVENTION

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When $P(A) > 0$, the conditional probability of B, given A is:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

- Rearrange this formula for $P(A \cap B)$

Similarly, when $P(B) > 0$:

$$P(A|B) = \frac{P(B \cap A)}{P(B)}$$

- What pieces of a Venn diagram would these correspond to?

Conditional probability JUUL and vaping example

Among those who have seen an ad for JUUL, what percent vaped in the past month?

$$P(\text{Vape} | \text{JUUL})$$

$$P(V|J) = \frac{P(J \cap V)}{P(J)}$$

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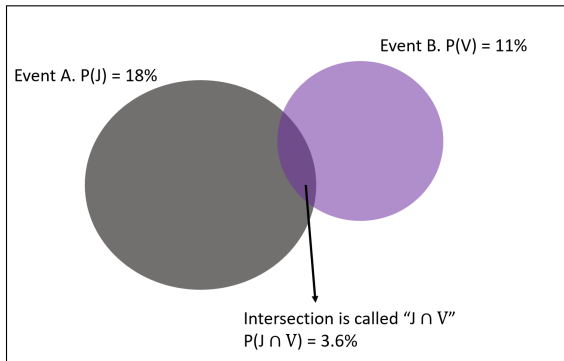
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Conditional probability

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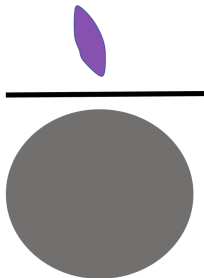
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What percent of individuals vape given
they have seen an ad for JUUL?



Conditional probability

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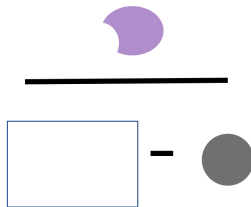
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$$P(V|J') = \frac{P(J' \cap V)}{P(J')}$$

What percent of individuals vape given
they have NOT seen an ad for JUUL?



Recall the definition for disjoint events

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- ▶ What did it mean for two events to be disjoint?
- ▶ Another term for disjoint is **mutually exclusive**
- ▶ How would we draw disjoint events in a Venn diagram?

Disjoint events

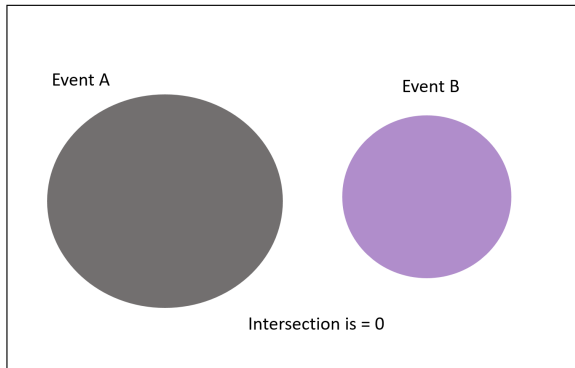
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Independent vs. dependent events

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What sorts of events are independent?

Two events are independent if knowing that one event occurred does not change the probability that the other occurred

Independent vs dependent: example 1. Down syndrome

Down syndrome is a genetic disorder caused when abnormal cell division results in an extra full or partial copy of chromosome 21.¹ The largest risk factors for having a child with Down syndrome are advanced maternal age.¹ Suppose that Martha is 40 and her baby has been diagnosed with Down syndrome. Martha's best friend Jane, also 40, is hoping to conceive. Is her baby's risk of Down syndrome independent of Martha's baby's risk?

1. <https://www.mayoclinic.org/diseases-conditions/down-syndrome/symptoms-causes/syc-20355977>

Independent vs. dependent events

Written out in probability notation, for any two events A and B, the events are independent if:

$$P(A|B) = P(A)$$

or

$$P(B|A) = P(B)$$

or

$$P(A \cap B) = P(A) * P(B)$$

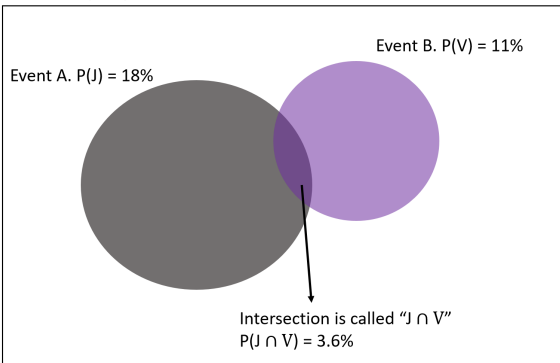
The “|” is read as “given” or “conditional on”

If one of these is true, they are all true.

Independent vs. dependent

A basic Venn diagram

Entire sample space S



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In our example, are seeing an ad for JUUL and vaping in the past month independent?

Independent vs. dependent events

From our JUUL and vaping example:

$$P(V|J) = \frac{P(V \cap J)}{P(J)} = \frac{.036}{.18} = .20$$

and

$$P(V) = 0.11$$

so

$$P(V|J) \neq P(V)$$

proving that we have events that are NOT independent: knowing someone's exposure to JUUL ads gives us information about their probability of having vaped.

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Multiplication rules

Multiplication rule

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For any two events, the probability that both events occur is given by:

$$P(A \cap B) = P(B|A) \times P(A)$$

Multiplication rule for independent events

Two events A and B are **independent** if knowing that one occurs does not change the probability that the other occurs.

If events are independent, $P(B|A) = P(B)$ so the general multiplication rule:

$$P(A \cap B) = P(B|A) * P(A)$$

Simplifies to:

$$P(A \cap B) = P(A) \times P(B)$$

Conditional Probability using tables

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Ad exposure	Vape	No Vape	Total
seen an ad for JUUL	0.036	0.144	0.18
did not see ad for JUUL	0.074	0.746	0.82
Total	0.11	0.89	1

What are the conditional probabilities of vaping by Ad exposure?

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Conditional probability and diagnostic testing

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Key terms:

- ▶ Sensitivity: $P(\text{test positive} \mid \text{truly have disease})$
- ▶ Specificity: $P(\text{test negative} \mid \text{truly do not have disease})$
- ▶ Positive predictive value: $P(\text{truly have disease} \mid \text{test positive})$
- ▶ Negative predictive value: $P(\text{truly do not have disease} \mid \text{test negative})$

Conditional probability and diagnostic testing

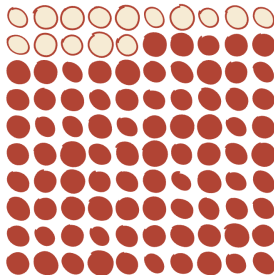
More key terms:

	Have the disease	Do not have the disease
Test positive	True positive	False positive
Test negative	False negative	True negative

The New York Times

When They Warn of Rare Disorders, These Prenatal Tests Are Usually Wrong

Some of the tests look for missing snippets
of chromosomes. For every 15 times they
correctly find a problem ○ ...



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Statistics is everywhere: prenatal testing

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“The tests initially looked for Down syndrome and worked very well. But as manufacturers tried to outsell each other, they began offering additional screenings for increasingly rare conditions. . . Nonetheless, on product brochures and test result sheets, companies describe the tests to pregnant women and their doctors as near certain. They advertise their findings as “reliable” and “highly accurate,” offering “total confidence” and “peace of mind” for patients who want to know as much as possible.”

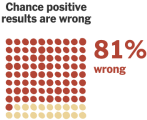
Statistics is everywhere: prenatal testing

THE UPSHOT | When They Warn of Rare Disorders, These Prenatal Tests Are Usually Wrong

As prenatal tests have expanded to more rare conditions, a larger share of their positive results are incorrect. Some of the worst-performing tests look for microdeletions, which are small missing snip-
AIBE-D-22-00448.pdf S.

DiGeorge syndrome

Affects 1 in 4,000 births
Can cause heart defects and delayed language acquisition.
(May appear on lab reports as “22q.”)



1p36 deletion

1 in 5,000 births
Can cause seizures, low muscle tone and intellectual disability.



Cri-du-chat syndrome

1 in 15,000 births
Can cause difficulty walking and delayed speech development.



Wolf-Hirschhorn syndrome

1 in 20,000 births
Can cause seizures, growth delays and intellectual disability.



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How to calculate the chance of having condition if you test positive?

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- ▶ The article gives the probability of NOT having the disease given that the test is positive, or the false positive proportion - for DiGeorge syndrome this is given as 81%.
- ▶ We don't have all of the information needed in the article to calculate all of the conditional probabilities, but if we imagine that the test has a 96% sensitivity and a 99.9% specificity we can look at how we could get to such a high proportion of false positives

Example: How to calculate the chance of having DiGeorge syndrome if you test positive?

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	Have DiGeorge	Do not have DiGeorge	Total
Test positive	True positive	False positive	
Test negative	False negative	True negative	
Total			

Example: How to calculate the chance of having DiGeorge syndrome if you test positive?

- ▶ 96% chance of testing positive when the individual has it (sensitivity)
- ▶ 99.9% chance of testing negative when the patient does not have it (specificity)
- ▶ 1 in 4000 births have this syndrome (0.025%) - this is one of the least rare conditions mentioned in the article

	Have DiGeorge	Do not have DiGeorge	Total
Test positive	24	100	124
Test negative	1	99875	99876
Total	25	99975	100000

Note that I have rounded the numbers in the non-disease column slightly

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What is the chance that a patient has DiGeorge syndrome given that they test positive?

- ▶ Positive predictive value: $P(\text{truly have disease} \mid \text{test positive})$
- ▶ $P(\text{truly have disease} \mid \text{test positive}) = 24/124 = 19.4\%$
- ▶ $P(\text{do not have disease} \mid \text{test positive}) = 100/124 = 80.6\%$

Parting humor

from XKCD

